TEXORA · LECTURE SHEET

Calculation of Dyes and Auxiliaries Chemicals

Every wet-process recipe is finally a set of multiplications. This sheet fixes the three bases — % owf, g/L and pick-up — and the conversions between them that production calculations depend on.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 2 · SPECIALIZATION — WET PROCESSING

1. The three dosing bases

- **% owf (on weight of fibre/fabric):** grams of product per 100 g of dry goods. Used for dyes and for chemicals whose action is on the goods. Independent of bath volume.
- **g/L (grams per litre of bath):** concentration in the liquor. Used for salt, alkali and auxiliaries that work in the bath, and for all padding liquors. Meaningless unless the liquor volume or ratio is stated.
- **Pick-up based (continuous processes):** the fabric carries away a measured volume of liquor per kilogram — the wet pick-up — so the bath concentration and the pick-up together set the amount deposited on the goods.

The **liquor ratio (LR)** links the first two bases: LR 1:R means R litres of bath per kilogram of fabric.

$$\text{Bath volume (L)} = \text{fabric mass (kg)} \times \text{LR} \;\mid\; \text{Dye (g)} = \%\text{ owf} \times \text{fabric mass (g)} / 100 \;\mid\; \text{Chemical (g)} = \text{conc. (g/L)} \times \text{bath volume (L)}$$

Conversion bridge: a chemical at C g/L in a bath of LR 1:R delivers C × R grams per kilogram of fabric, i.e. C × R / 10 % owf. Conversely, a dose of P % owf corresponds to P × 10 / R g/L in that bath. Every cross-check in this sheet uses this bridge.

2. Fully worked bulk calculation — jet dyeing of cotton knit

WORKED EXAMPLE 1 — 500 KG COTTON KNIT, JET MACHINE, LR 1:8, REACTIVE DYES

Bath volume = 500 × 8 = **4,000 L**.

Dyes, total 2.35% owf (trichromacy): dye mass = 2.35 × 500,000/100 = **11.75 kg**.

Glauber's salt 60 g/L → 60 × 4,000 = 240,000 g = **240 kg** (bridge check: 60 × 8/10 = 48% owf —

salt is correctly bath-dosed, never quoted as % owf).

Soda ash 15 g/L → 15 × 4,000 = **60 kg**. Wetting agent 1.5 g/L → **6.0 kg**. Sequestering agent 2 g/L → **8.0 kg**.

Total chemical load to effluent per batch ≈ 240 + 60 + 6 + 8 = 314 kg plus unfixed dye — the reason recipe arithmetic is also effluent arithmetic.

Note the discipline: dyes by mass of goods, everything in the bath by bath volume. Mixing the two bases — ordering salt as "% owf" from a recipe written in g/L — is the commonest store-room error, and it scales silently with LR changes.

3. Laboratory scale and stock solutions

Laboratory lots of 5–10 g cannot be dosed by weighing milligrams. Dyes and liquid chemicals are dispensed from **stock solutions** of known concentration: a 1% w/v dye stock contains 10 g/L = 10 mg/mL; volume to pipette (mL) = dye required (mg) / stock strength (mg/mL).

WORKED EXAMPLE 2 — LAB LOT FROM STOCK SOLUTIONS

Lot: 10 g cotton, LR 1:20 → bath = 200 mL. Recipe: dye 1.50% owf; salt 40 g/L; soda ash 10 g/L.

Dye = 1.50% × 10 g = 0.150 g = 150 mg → from 1% stock (10 mg/mL): 150/10 = **15.0 mL**.

Salt = 40 × 0.200 = **8.0 g**; soda ash = 10 × 0.200 = **2.0 g**.

Make the bath up to the final 200 mL *after* all pipetted additions, or the stock volumes dilute every g/L figure.

4. Continuous (pad) calculations — pick-up does the dosing

In padding there is no bath equilibrium: the goods leave with a fixed liquor volume. If wet pick-up is W (as a fraction, e.g. 0.70), one kilogram of fabric carries away W litres of liquor, so:

Amount on fabric (g/kg) = bath conc. (g/L) × W | Required bath conc. = target (g/kg) / W

WORKED EXAMPLE 3 — PAD BATH CONCENTRATION FOR A TARGET DEPTH

Target: 2.0% owf of dye fixed on the fabric = 20 g per kg. Wet pick-up 70% → each kg carries 0.70 L.

Required bath concentration = 20 / 0.70 = **28.6 g/L**.

Check: 28.6 g/L × 0.70 L/kg = 20.0 g/kg ✓. At 60% pick-up the same target would need 33.3 g/L — pick-up is a recipe variable, not a machine detail.

5. Recipe costing

Cost per kilogram of fabric = Σ (quantity per kg) × (price per unit). For Worked Example 1: dye 11.75 kg/500 kg = 23.5 g/kg of fabric; salt 240/500 = 0.48 kg/kg. Even without prices, the mass table exposes the truth of wet processing: the salt outweighs the dye twenty to one, which is why low-LR machinery (Topic 11) and salt-recovery thinking dominate modern dyehouse economics. When quotations are compared, compare the full table — dyes, salt, alkali, auxiliaries, water and steam — never the dye line alone.

6. Common calculation faults

FAULT	CONSEQUENCE
Quoting bath chemicals in % owf	Dose changes silently when LR changes between lab and bulk; shade and levelness drift.
Forgetting stock volume in lab baths	All g/L concentrations diluted; lab shade lighter than the recipe predicts.
Using wet fabric mass	% owf computed on water as well as fibre; systematic under-dosing. Recipes are always on dry goods mass.
Ignoring pick-up in pad recipes	Bath made to the on-fabric target as g/L; actual deposit short by the pick-up fraction.
Rounding intermediate values	Errors compound across a trichromacy; carry full precision and round only the final weighed quantities.

7. Glossary — key terms

- **Liquor ratio:** litres of bath per kilogram of goods, written 1:R.
- **Wet pick-up:** liquor retained by fabric after padding, expressed as % of dry mass or L/kg.
- **Stock solution:** a standardised solution from which small quantities are dispensed volumetrically.
- **Auxiliaries:** processing chemicals other than dyes — wetting, sequestering, levelling agents, salt and alkali.

8. Key points

- Dyes are dosed on goods mass (% owf); bath chemicals on bath volume (g/L); the bridge C × R/10 converts between them.
- In padding, bath concentration × pick-up = deposit on fabric; the recipe is incomplete without the pick-up.

- Every recipe should survive a bridge cross-check and a full mass table before it is costed or issued.

9. Summary

Dyehouse arithmetic has only three ideas — mass of goods, volume of bath, and the pick-up that links them in continuous work — but they must never be confused. The worked examples in this sheet (4,000 L bath, 240 kg salt against 11.75 kg dye; 28.6 g/L in the pad bath for a 2% deposit) are the template for every calculation in the topics that follow.

10. Practice

1. Bulk lot: 800 kg fabric, LR 1:6, dye 1.85% owf, salt 45 g/L, soda ash 12 g/L. Calculate bath volume and all three quantities.
2. Lab lot: 5 g, LR 1:30, dye 0.80% owf from a 0.5% w/v stock. How many mL of stock are pipetted?
3. A pad recipe must deposit 45 g/kg of a finishing agent at 75% pick-up. What bath concentration is required?
4. Convert 25 g/L at LR 1:10 to % owf, and 6% owf at LR 1:15 to g/L.

Answer key

1. Bath = 800 × 6 = 4,800 L. Dye = 1.85% × 800 kg = 14.8 kg. Salt = 45 × 4,800 = 216 kg. Soda ash = 12 × 4,800 = 57.6 kg.
2. Dye = 0.80% × 5 g = 0.040 g = 40 mg. 0.5% stock = 5 mg/mL → 40/5 = 8.0 mL.
3. Conc. = 45/0.75 = 60 g/L.
4. 25 g/L at 1:10 → 25 × 10/10 = 25% owf. 6% owf at 1:15 → 6 × 10/15 = 4.0 g/L.

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